KONERU LAKSHMAIAH EDUCATION FOUNDATION

Department of Freshman Engineering / Department of ECE

PROJECT DOCUMENTATION

4-Bit Binary to Decimal Converter Using Logic Gates

Course Code: 26EC1202E
Capstone Digital Logic Project
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Project Guide
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Abstract

This project presents the design of a 4-bit binary-to-decimal converter using combinational logic gates and BCD-to-7-segment decoder/driver ICs. A 4-bit input represents 16 possible values, from 0000_2 to 1111_2, corresponding to decimal values 0 through 15. The circuit separates the output into tens and units digits and displays the result on two common-anode 7-segment displays. Boolean expressions and Karnaugh-map minimization are used to derive the logic, and the design is considered for verification in Logisim Evolution and implementation using 74xx TTL components. The project demonstrates how binary values can be converted into a human-readable decimal display without relying on a microcontroller or firmware.

1. Introduction

Digital systems store and process information in binary form, using only 0 and 1. Although binary is suitable for digital circuits, decimal numbers are easier for people to read. A binary-to-decimal converter bridges this gap by translating a binary input into decimal digits for display.

This project focuses on a 4-bit input. Four bits provide 2^4 = 16 combinations, ranging from 0000_2 (decimal 0) to 1111_2 (decimal 15). Since the output may contain two decimal digits, the circuit uses separate tens and units outputs, followed by decoder/driver circuits and 7-segment displays.

2. Problem Statement

Design a combinational digital circuit that accepts a 4-bit binary input and displays its equivalent decimal value from 00 to 15 on two 7-segment displays. The design should correctly handle the transition from decimal 9 to decimal 10 and should avoid invalid hexadecimal-style symbols for input values 10-15.

3. Objectives

1. Convert all 16 four-bit binary input combinations into decimal display values from 00 to 15.

2. Construct a truth table for the tens and units outputs.

3. Derive and simplify Boolean expressions using Boolean algebra and Karnaugh maps.

4. Interface the logic outputs with 7447 BCD-to-7-segment decoder/driver ICs and common-anode displays.

5. Test the design through simulation and, where available, breadboard implementation.

4. Background Theory

A 4-bit binary number has positional weights 8, 4, 2, and 1. Its value is B = 8B_3 + 4B_2 + 2B_1 + B_0. For values 0-9, the tens digit is 0 and the units digit is the same numerical value. For values 10-15, the tens digit is 1 and the units digit is the input value minus 10.

A 7-segment display contains seven LED segments, conventionally labelled a through g. A 7447 decoder/driver accepts a BCD input and drives a common-anode display. The converter logic must therefore produce valid BCD digits for the tens and units positions before they reach the display drivers.

5. Components Required

• IC 7408 - quad 2-input AND gates (quantity: 1).

• IC 7432 - quad 2-input OR gates (quantity: 1).

• IC 7404 - hex inverter (quantity: 1).

• IC 7447 - BCD-to-7-segment decoder/driver (quantity: 2).

• IC 7474 - dual D flip-flop, listed in the presentation for input synchronization (quantity: 1).

• Two common-anode 7-segment displays.

• One 4-position DIP switch for inputs B_3, B_2, B_1, and B_0.

• Four 10 kohms pull-down resistors and fourteen 330 ohms current-limiting resistors.

• Regulated +5 V DC supply, breadboard, and connecting wires.

6. System Architecture

Signal flow: 4-bit DIP-switch input -> combinational logic converter -> tens and units BCD outputs -> two 7447 decoder/drivers -> two common-anode 7-segment displays.

The converter logic identifies whether the input is below 10 or at least 10. It generates the tens digit (0 or 1) and the corresponding units digit (0-9). Each BCD digit is then sent to a 7447 decoder/driver, which controls the segments of its display.

4-bit DIP Switch Input

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Combinational Logic Converter

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Tens BCD (0/1) + Units BCD (0-9)

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Two 7447 Decoder/Drivers

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Two Common-Anode 7-Segment Displays

7. Circuit Working Principle

For binary inputs 0000 through 1001 (decimal 0-9), the tens digit is 0 and the units digit represents the same decimal value. For inputs 1010 through 1111 (decimal 10-15), the tens digit is 1 and the units digit represents decimal 0 through 5, respectively.

The outputs from the converter are applied to the two 7447 decoder/drivers. The drivers control the common-anode displays through current-limiting resistors. The presentation also lists a 7474 D flip-flop for input synchronization; if used, its role should be confirmed against the final circuit schematic.

8. Truth Table

Table 1 lists the intended decimal mapping for all 16 input combinations. T is the tens digit; U_3U_2U_1U_0 is the BCD units digit.

Decimal	Input B_3B_2B_1B_0	Tens T	Units U_3U_2U_1U_0	Display
0	0000	0	0000	00
1	0001	0	0001	01
2	0010	0	0010	02
3	0011	0	0011	03
4	0100	0	0100	04
5	0101	0	0101	05
6	0110	0	0110	06
7	0111	0	0111	07
8	1000	0	1000	08
9	1001	0	1001	09
10	1010	1	0000	10
11	1011	1	0001	11
12	1100	1	0010	12
13	1101	1	0011	13
14	1110	1	0100	14
15	1111	1	0101	15

Note: The BCD units output for decimal 8 and 9 includes U_3 = 1.

9. Boolean Equations

The presentation gives the following minimized expressions:

$$T = B_3B_2 + B_3B_1 = B_3(B_2 + B_1)$$

$$U_2 = B_2B_3' + B_2B_1 = B_2(B_3' + B_1)$$

$$U_1 = B_3'B_1 + B_3B_2B_1'$$

$$U_0 = B_0$$

Important verification note: the presentation also states U_3 = 0 always, but its own truth table shows U_3 = 1 for decimal 8 and 9 (inputs 1000 and 1001). To match the truth table and intended BCD output, U_3 should be high for 8 and 9 and low for 10-15. A suitable expression is U_3 = B_3·T', where T is the tens output. Please confirm this equation against your final K-map and circuit before submission.

10. Simulation and Hardware Testing

The presentation describes building a modular converter in Logisim Evolution and applying test vectors for inputs 0000-1111. The stated checks include verifying the segment mapping a-g, checking the 9-to-10 boundary transition, and confirming outputs 00 through 15.

For physical implementation, the listed TTL ICs are placed on a breadboard, powered from a regulated 5 V supply, and connected to the common-anode displays through 330 ohms current-limiting resistors. Each input combination should be tested against the truth table. Record actual observations from your own simulation or hardware test before submitting this section as completed experimental work.

11. Results and Discussion

The intended result is a two-digit decimal display covering the complete input range from 00 to 15. The truth table provides the expected output for every four-bit input. Correct operation depends on generating valid BCD tens and units digits and connecting the 7447 drivers to the displays with the correct polarity and segment wiring.

12. Advantages

• Fast conversion using combinational logic, with no firmware execution cycle required.

• Uses commonly available 74xx TTL logic ICs.

• Demonstrates binary number systems, Boolean algebra, Karnaugh maps, decoders, and display interfacing.

• Provides a visible way to inspect 4-bit digital values.

13. Limitations

• The circuit supports only four-bit inputs, covering values 0-15.

• A gate-level design requires multiple IC connections, resistors, and breadboard jumpers.

• Mechanical switch bounce may cause unstable readings while changing inputs; debouncing or registered inputs may be needed.

• The final output depends on correct Boolean equations, wiring, and 7-segment decoder connections.

14. Applications and Future Scope

Possible applications include educational demonstrations, digital instrumentation, and visual monitoring of small binary values in digital systems.

Future improvements may include extending the converter to 8-bit inputs using a suitable conversion architecture, implementing the logic in Verilog HDL for FPGA/ASIC synthesis, and using display multiplexing to reduce hardware connections. Input debouncing or synchronization can also be added where needed.

15. Conclusion

The 4-bit binary-to-decimal converter illustrates how combinational logic can translate binary values into decimal digits for a human-readable display. The design uses a truth table, Boolean expressions, logic gates, BCD decoder/drivers, and 7-segment displays to represent values from 00 to 15. Simulation and hardware testing can be used to verify each input combination. Before final submission, the U_3 equation should be reconciled with the truth table, as noted in the Boolean-equations section.

References

1. Project presentation: "4-Bit Binary to Decimal Converter Using Logic Gates," Course Code 26EC1202E, Academic Year 2026-27.

2. Datasheets for the 7404, 7408, 7432, 7447, and 7474 integrated circuits, as applicable to the components used in the final circuit.